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Evaluating genetic differentiation of *Anopheles arabiensis* in relation to larval habitats in Kenya

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Abstract

Anopheles arabiensis, an important vector of malaria in sub-Saharan Africa shows great versatility in utilizing a broad range of larval habitats. Most studies on the population genetic structure of this species have focused on adult mosquitoes collected mainly indoors and have found populations within relatively short distances to be panmictic. Various authors suggested that the incipient speciation between the chromosomal forms of *Anopheles gambiae* s.s. in West Africa involved adaptation to different habitats of larvae. To investigate if similar process occurs in *An. arabiensis*, we analyzed samples from four different larval habitats alongside a sample of adult mosquitoes collected indoors at eight microsatellite loci. We tested the hypothesis that *An. arabiensis* is subdivided into populations that are adapted to exploit different larval habitats. Overall, the level of genetic differentiation among the larval subpopulations was low with the mean pairwise Fst estimate based on information from all the loci being 0.0232 ($P = 0.1691$) although differentiation was only significant for comparisons between certain larval samples and the adult samples. In addition, there was strong evidence against subdivision among larvae when multilocus simultaneous assignment tests were used. These results do not provide sufficient support for the hypothesis that *An. arabiensis* is subdivided into subpopulations that specialize to exploit different larval habitats.
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Keywords: *Anopheles arabiensis*; Larval sites; Gene flow; Microsatellite loci; Kenya

1. Introduction

Malaria continues to be an important health problem in the tropics, with an estimated 350–500 million clinical cases occurring annually and 60% of these being found in sub-Saharan Africa (WHO, 2005). Two members of the *Anopheles gambiae* species complex, namely *Anopheles gambiae* s.s. and *Anopheles arabiensis*, together with *Anopheles funestus* are the most important vectors of malaria in tropical Africa (Gilles and De Meillon, 1968). The recognition of existing heterogeneities in vector populations is important for the understanding of the epidemiology of malaria because different vectors display considerable variation in their biology and in behavioural traits that impact on vector potential (Gilles and De Meillon, 1968). Further, information on the level of genetic isolation between populations is of relevance in the implementation of malaria control measures based on population replacement through transgenic technology.

Anopheles arabiensis, though often found in sympatry with *An. gambiae* s.s., shows incredible versatility, exemplified by the ability to utilize both human and non-human blood meal sources depending on availability and a broader range of larval sites compared to other members of the species complex (White, 1974). Notable is the presence of this species in areas of rice irrigation agriculture in East Africa where *An. gambiae* s.s. is virtually non-existent (Githeko et al., 1990; Ijumba et al., 1990). Studies of genetic variability based on chromosomal inversions have revealed high levels of complexities in West African populations of *An. gambiae* s.s. and *An. funestus* but virtually none in *An. arabiensis*. These studies have found associations between specific inversions and feeding and resting behaviour as well as crepuscularity and adaptation to

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different ecological clines (Coluzzi et al., 1979; Petrarca and Beier, 1992; Powell et al., 1999; Toure et al., 1998; Cohuet et al., 2005). High degrees of reproductive isolation have even been inferred between the chromosomal forms, suggesting the occurrence of adaptive evolution (Cohuet et al., 2005; Toure et al., 1998). Many studies have assessed genetic differentiation in vectors of malaria to infer levels of gene flow. Most of these studies have focused on adult mosquito populations mainly collected resting indoors and have generally found populations occurring within short distances such as between houses or villages to be panmictic (e.g. Lehmann et al., 1997; Kamau et al., 1998a; Braginets et al., 2003; Temu and Yan, 2005). The finding in several studies of significant deviations from Hardy–Weinberg expectations within mosquito populations even when the smallest sampling unit, the house, is considered (Lehmann et al., 1997; Kamau et al., 1998a; Donnelly et al., 1999) may be due to pooling of genetically separate subpopulations, suggesting that a subdivision exists, possibly based on ability of mosquitoes to utilize certain larval habitats. Studies in West Africa have found associations between the occurrence of specific chromosomal forms and climatic differences as well as larval breeding habitats types (Toure et al., 1994, 1998). We tested the hypothesis that the ability to utilize a wide range of larval habitats by *An. arabiensis* mosquitoes is a consequence of adaptation of different subpopulations to different habitats and that this has resulted in restricted gene flow between these subpopulations. Genetic differentiation at microsatellite loci was estimated in larval mosquitoes from four different types of larval habitats namely hoof prints, streamside pools, rice paddies and tyre ruts, and adult mosquitoes found resting indoors in the same area. The four types of larval habitats that were sampled differ in characteristic such as size, duration, presence of pesticides and nutrients.

2. Materials and methods

2.1. Mosquito collection, preservation and sampling scheme

This study was carried out in Mwea area (00°67′S, 37°35′E) of Central Kenya, a rice-growing area which has traditionally had a single growing season extending from June to December. In the last 3 years or so, however, as a result of the prevailing drought in the country and the associated water shortages, different sections of the area are flooded intermittently thus maintaining almost all-year-round rice growing but with June–December being the main growing season. Specimen were collected both as larvae from rice paddies using standard dippers and pipettes for smaller larval sites and as adults by aspiration from walls inside human dwellings. Collections of the adult mosquitoes and most of the larvae were made on 4th and 5th August 2004 and additional sampling of larvae carried out on 9th and 10th September 2004 to boost sample sizes. All the four types of larval sites were sampled at each sampling time point. Sampling was during the dry season, which coincided with the main rice growing season and most paddies were flooded. Specimens were identified as *An. gambiae* s.l. based on morphological characteristics (Gilles and De Meillon, 1968). Larvae from the different paddies were preserved live in separate bottles and transported to the laboratory. All specimens were desiccated and preserved dry until further analyses. To reduce the chances of including siblings which are genetically closely related in our analysis of genetic variability, multiple larval sites were sampled for each of the different types of larval habitats selected and only one specimen included in our analysis. Thus, for example, a sample of 40 larvae for a particular larval habitat type was constituted by sampling 40 different sites of the same type. All larval habitats sampled were within a 3 km radius with the distance between habitats being as little as 10 cm in the case of hoof prints.

2.2. DNA extraction and species identification

DNA from individual specimens was extracted by the alcohol precipitation method of Collins et al. (1987) and re-suspended in sterile distilled water. Specimens were identified further to sibling species of the *An. gambiae* complex using the species-specific polymerase chain reaction assay of Scott et al. (1993).

2.3. Microsatellite analysis

We used a total of eight microsatellite loci that were cloned from *An. gambiae* s.s. as described in Zheng et al. (1996) and Kamau et al. (1998a). These loci were selected after an initial screening process in which DNA from ten specimens was amplified at the loci and amplification found to be satisfactory and the loci to have least two alleles. In addition, all loci used were located outside chromosomal inversion regions of *An. arabiensis* as location of microsatellite loci within these inversions has been shown to affect the level of differentiation that they measure (e.g. Kamau et al., 1998b; Lanzaro et al., 1998; Temu and Yan, 2005). Standard PCR reactions were run in 15 μl reactions in a Gene Amp 9700 system and alleles scored against allelic ladders following electrophoresis and silver staining as previously described (Kamau et al., 1998a).

2.4. Data analysis

Differences in the mean number of alleles and mean heterozygosities between subpopulations were tested using analysis of variance (ANOVA) test. Deviations from Hardy–Weinberg expectations and their statistical significance were assessed using exact tests available in Genepop 3.1b. (Raymond and Rousset, 1995) and the significance of multiple tests assessed using Fisher's exact test. Differentiation was estimated by Fst (the variance in allele frequencies standardized by mean allele frequency among samples) to determine the level of genetic variability among samples. The significance of Fst values was evaluated using exact tests of genotypic differentiation (Goudet et al., 1996) in Genepop. We also evaluated population structure among the larval collections using a Bayesian statistical model as implemented by the program

STRUCTURE (Pritchard et al., 2000). The frequency of null alleles (r) at the locus that did not conform to Hardy–Weinberg expectations was estimated using the formula: $r = (H_E - H_o)/(H_E + H_o)$ according to Chakraborty et al. (1992).

3. Results

The mean number of alleles across all loci was moderate (range 3.9–5.1) and did not differ significantly between the five subpopulations (ANOVA, $F = 0.4178$, d.f. = 4, 35, $P = 0.7946$). The allele frequency distribution was very similar across all the five subpopulations with one allele being the most common in all the subpopulations for all loci studied except loci AGXH180, AG2H46 and AG3H577. Even for these three loci, one of two alleles was the most common in all the subpopulations. Mean heterozygosity across all loci was 0.40 (S.E. 0.08), with the locus-specific range of 0.16–0.67. Mean heterozygosities between loci were significantly different from each other (ANOVA, $F = 12.7640$, d.f. = 7, 30, $P < 0.0001$) but not between populations (ANOVA, $F = 0.3107$, d.f. = 4, 33, $P = 0.8687$).

Significant deviations from Hardy–Weinberg expectations were detected at locus AGXH180 for all the subpopulations tested ($P < 0.01$, Table 1). Excluding this locus, overall

Table 1
Genetic variability in *An. arabiensis* for the four types of larval habitats and adults indoor-resting mosquitoes

Locus (cyt/repeat[a])	Statistic	Rice paddies	Hoof prints	Streamside pools	Tyre ruts	Adults (indoors)
AGXH180 (X/GT)	N[b] (A#)[c]	82 (9)	36 (8)	50 (7)	42 (10)	92 (9)
	Common allele%	32.9	44.4	28.0	23.8	23.9
	He (Ho)	0.81 (0.54)	0.77 (0.39)	0.83 (0.48)	0.88 (0.43)	0.85 (0.46)
	Fis	+0.337**	+0.494**	+0.422**	+0.506**	+0.347**
1D1 (X/CCA)	N (A#)	78 (2)	52 (2)	60 (1)	54 (2)	104 (2)
	Common allele%	96.2	96.2	100.0	94.4	99.0
	He (Ho)	0.10 (0.08)	0.11 (0.08)	–	0.11 (0.11)	0.04 (0.02)
	Fis	−0.027ns	−0.020ns	–	−0.040ns	–
AG2H46 (II/GT)	N (A#)	74 (6)	54 (5)	64 (5)	50 (5)	82 (5)
	Common allele%	39.2	37.0	50.0	50.0	34.1
	He (Ho)	0.69 (0.62)	0.69 (0.78)	0.60 (0.69)	0.63 (0.60)	0.70 (0.66)
	Fis	+0.087ns	−0.125*	−0.148ns	+0.043ns	+0.042ns
22C1 (II/TG)	N (A#)	78 (7)	34 (5)	54 (4)	46 (7)	84 (6)
	Common allele%	39.7	35.3	24.1	30.4	38.1
	He (Ho)	0.74 (0.51)	0.71 (0.71)	0.71 (0.63)	0.74 (0.78)	0.62 (0.62)
	Fis	+0.304**	+0.003*	+0.090ns	−0.052ns	0.000ns
29C (III/TGA)	N (A#)	82 (4)	38 (2)	60 (2)	46 (2)	70 (1)
	Common allele%	87.8	92.1	93.3	89.1	100.0
	He (Ho)	0.24 (0.15)	0.20 (0.16)	0.16 (0.13)	0.24 (0.22)	–
	Fis	+0.349*	−0.059ns	−0.055ns	−0.100ns	–
AG3119 (III/GT)	N (A#)	68 (4)	52 (3)	64 (4)	62 (4)	70 (3)
	Common allele%	89.7	84.6	76.6	80.6	91.4
	He (Ho)	0.19 (0.18)	0.30 (0.31)	0.40 (0.47)	0.34 (0.29)	0.19 (0.17)
	Fis	+0.083ns	−0.140ns	−0.176ns	+0.151*	−0.065ns
AG3H577 (III/GT)	N (A#)	84 (5)	44 (4)	56 (5)	46 (5)	88 (5)
	Common allele%	54.8	52.3	48.2	45.7	46.6
	He (Ho)	0.58 (0.81)	0.61 (0.41)	0.65 (0.50)	0.68 (0.70)	0.70 (0.68)
	Fis	−0.395**	+0.336*	+0.220ns	−0.045ns	+0.002**
45C (III/GA)	N (A#)	88 (4)	60 (3)	58 (3)	70 (3)	90 (4)
	Common allele%	86.4	91.7	86.2	85.7	86.7
	He (Ho)	0.27 (0.23)	0.19 (0.17)	0.28 (0.28)	0.26 (0.26)	0.26 (0.22)
	Fis	+0.088ns	−0.058ns	−0.098ns	+0.005ns	+0.097ns
Mean	N (A#)	79.3 (5.12)	46.3 (4.00)	58.3 (3.88)	52.0 (4.75)	85.0 (4.38)
	Common allele%	65.84	67.07	63.30	62.46	64.97
	He (Ho)	0.45 (0.47)	0.45 (0.37)	0.52 (0.36)	0.48 (0.41)	0.48 (0.46)
	Fis	0.1214*	0.1020ns	0.0835ns	0.0984ns	0.1418ns

The table shows cytological location of loci and their repeat motif, sample sizes, number of alleles per locus, percentage of the most common allele, the observed (Ho) and expected heterozygosities (He) and Fis values.

[a] cyt/repeat: cytological location of loci/repeat motif.

[b] N: total number of alleles scored.

[c] A#: number of alleles at locus.

* $P < 0.01$.

** $P < 0.05$.

ns $P > 0.05$.

deviations from Hardy–Weinberg expectations were not significant (Hardy–Weinberg global test, $P = 0.1431$). The mean frequency of null alleles at locus AGXH180 across all subpopulations was 0.2031 ± 0.0146 (range 0.1552–0.2416). Locus AGXH180 was, therefore, excluded from the estimation of genetic variability to avoid the possible bias on the estimates of genetic variability that the presence of null alleles might cause.

Overall, the level of genetic differentiation as measured by Fst was low, with the mean Fst estimate based on the information from all the loci for all the larval subpopulations being 0.0232 ($P = 0.1691$) (Locus-specific Fst range: 0.0000–0.1257). Pairwise comparisons of the larval subpopulations also revealed low levels of differentiation, with the mean Fst based on information from all loci ranging from 0.0000 to 0.0423 and only 2 out of a total of 42 pairwise Fst values being significant (2.1 significant values are expected by chance at the 0.05 significance level). However, when the adult mosquito sample was included in the pairwise comparison, a total of 12 out of 70 comparisons were significant (3.5 significant values are expected at the 0.05 significance level) with 10 of these being for comparisons between one of the four larval collections and the adult mosquitoes collected indoors. This suggests that, contrary to our hypothesis, the differentiation between the larval subpopulations contributed minimally to the differentiation found.

To test for the Wahlund effect, we pooled all the larval subpopulations into one population and evaluated the effect of the pooling on Hardy–Weinberg expectations. Significant deviation were observed in three of the seven loci (excluding locus AGXH180, Fisher's exact test: $\chi^2 = 35.4823$, d.f. = 14, $P = 0.0012$). Importantly, the mean Fis across the four larval subpopulations (0.061) was not significantly smaller than that of the pooled larval samples (0.079) when tested using paired t test across eight loci (difference = 0.018, $P > 0.2$, $t = 1.4$, d.f. = 7). This indicates that the observed deviations from Hardy–Weinberg expectations when the larval subpopulations were merged were mostly due to an increase in statistical power of the test as a result of the increased sample size rather than from Wahlund effect.

To determine if the larval specimens were divided into two or more populations, the data from the larval collections were analyzed using the software STRUCTURE which uses iterative computation process to simultaneously assign multilocus genotypes into populations and estimate the probability of observing the data, given the number of populations and their estimated allele frequencies. The probability of obtaining the data increases as departures from equilibrium within and between loci decrease, given the number of populations and their allele frequencies. We used the admixture model with correlated allele frequencies, which best applies to our situation (with 10^5 burn-in iterations, followed by 10^6 Markov Chain Monte Carlo iterations). The likelihood of obtaining these data decreased substantially from one population to two (means over three independent simulations: LnP(D) for $K = 1$ and 2 were -1712 and -1842, respectively) providing strong evidence against subdivision.

4. Discussion

We analyzed variability at microsatellite loci in *An. arabiensis* to test the hypothesis that it is subdivided into populations that specialized to exploit different larval habitats, resulting in genetic differentiation between them. This hypothesis stemmed from observations revealing sharp discontinuity in distribution of species and populations of members of the *An. gambiae* species complex in relations to larval habitats. In West Africa, the M molecular form of *An. gambiae* s.s. dominates near rice fields, whereas outside the rice fields, the S molecular form dominates (Toure et al., 1994; Coluzzi et al., 1979). In East Africa, a similar situation is observed in that whereas the M form of *An. gambiae* s.s. is absent, *An. arabiensis* dominates in rice fields with *An. gambiae* s.s. being virtually non-existent (Githeko et al., 1990; Ijumba et al., 1990). Various authors have suggested that the distribution of the different *An. gambiae* s.s. chromosomal forms reflects larval specialization to different habitats, possibly through the accumulation of adaptive alleles within chromosomal inversions leading to reproductive isolation and eventually to incipient speciation (Powell et al., 1999; Coluzzi et al., 2002). Because *An. arabiensis* exploits at least as diverse larval habitats as *An. gambiae* s.s., we predicted that adaptation to these habitats may have resulted in a degree of genetic isolation between the subpopulations.

Four different larval habitats namely, rice paddies, cattle hoof prints, streamside pools, and tyre ruts alongside adult female mosquitoes resting indoors were sampled. Rice paddies are characterized by being more permanent and larger than the other larval habitats and having pesticides and fertilizers used in rice cultivation. The hoof prints, representing the smallest larval sites, were close to cattle watering points and were unique in the sense that cow dung provides an additional source of particulate organic matter which is a major food resource for the larvae (Clements, 1992). Streamside ponds were generally shaded and more or less permanent, while tyre ruts were very transient and typically lasted about a week. Because larval sites were dispersed throughout the landscape, our sampling approach had the power to detect genetic isolation in a truly sympatric environment.

Overall, pairwise comparisons of larval subpopulations representing the different habitats yielded very low levels of genetic differentiation. In addition, there was strong evidence against population subdivision when genotypes were assigned to subpopulations without prior information on their origin using STRUCTURE. Thus, we conclude that despite slight variation between adult and larval samples, which may be attributable to sampling and genetic drift between generations, the gene pool of *Anopheles arabiensis* populations is not subdivided among larval habitats. It has been hypothesized that maintaining together co-adapted genes that confer the capacity to exploit different habitats requires restricted gene flow as is seen between the molecular forms of *An. gambiae* (della Torre et al., 2002; Coluzzi et al., 2002). The absence of genetic isolation between *An. arabiensis* samples from different larval habitats suggests that the capacity to exploit these distinct habitats does not involve restricted gene flow between

subpopulations that are adapted to different conditions. Instead, one population shows a remarkable capacity to exploit a wide range of distinct larval habitats.

These results agree with studies showing no spatial segregation between sympatric *An. gambiae* s.s. and *An. arabiensis* in Western Kenya (Gimnig et al., 2001) or between sympatric populations of the different *An. gambiae* s.s. molecular forms in West Africa (Edillo et al., 2002) as well as with studies showing no differential capacity to exploit different larval habitats by the molecular forms (Diabate et al., 2005). Nevertheless, all these studies fail to explain the segregation between molecular forms in West Africa between rice fields and the areas surrounding them and the segregation between *An. gambiae* s.s. and *An. arabiensis* occurring across these habitats. Finally, the lack of genetic isolation between *An. arabiensis* from different larval habitats is beneficial for malaria control through the introduction of refractoriness genes: such genes would be integrated into natural mosquito populations almost unhindered.

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